TeaGuard AI

Intelligent Decision Support System for Sri Lankan Tea Estate Management

R26-IT-118

Final Viva Presentation | Academic Year 2026

PP2 Feedback & Actions Taken

PP2 Comments (verbatim)	PP2 Panel Comment	Actions Taken
IT22209320 Miuranga W.A.R.	*"satisfactory"*	Improved the tea leaf disease detection model, user interface (UI), and backend functionality to enhance system usability and performance.
IT22279620 Gunasekara G.N.D.	*"satisfactory"*	Improved the weather and climate module's model, user interface (UI), and backend functionality.
IT22267986 Gunasekara L.M.N.P.	*"No idea about how the values are coming to the different fields. Not aware of her own component."*	Improved understanding of all component functions, data flow, field values, backend processing, and how the model-generated results are used in the system.
IT22282668 Pathirana I.M.	*"Very similar images again... Initially mentioned as taken from Google then as from a Tea-factory? Model predicts incorrect answer for new images (google)"*	Collected new tea-sorting image data, planned/undertook tea-factory data collection, and retrained the model to improve prediction reliability on unseen images.

Impact of the Problem

To the Estate:

- Late disease detection causes yield loss.
- Uneven workloads increase worker fatigue.
- Poor weather planning affects irrigation and harvesting.
- Manual grading causes inconsistent quality.

To the Industry:

- Decisions rely on manual records and experience
- Four problems handled in four disconnected ways

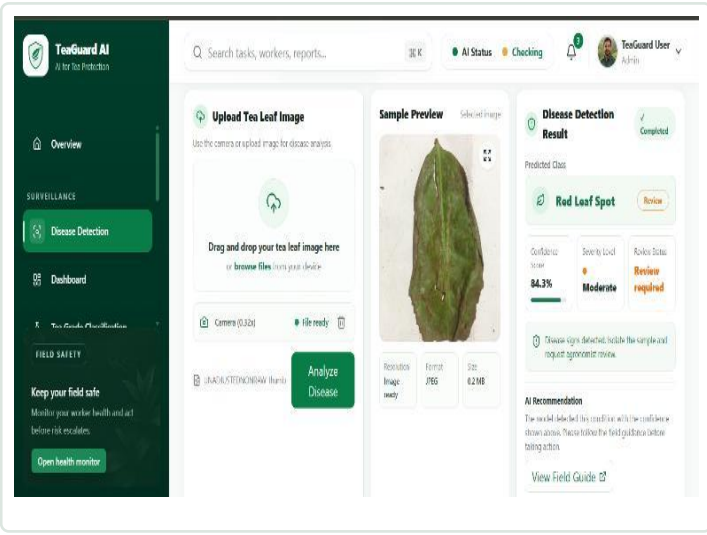
Existing Practice & Its Limitations

- **Manual inspection:** Slow and depends on inspector experience.
- **Paper records:** No real-time worker monitoring.
- **General weather forecasts:** Not linked to tea disease risks or actions.
- **Manual tea grading:** Subjective and inconsistent.

Tea Leaf Disease Detection & Pattern Analysis



- **Disease Detection:** Identifies tea leaf diseases from uploaded images.
- **Disease Classes:** Algal Spot, Brown Blight, Gray Blight, Helopeltis, Red Leaf Spot, and Healthy.
- **Output:** Shows the predicted class, confidence score, and treatment recommendations for user review

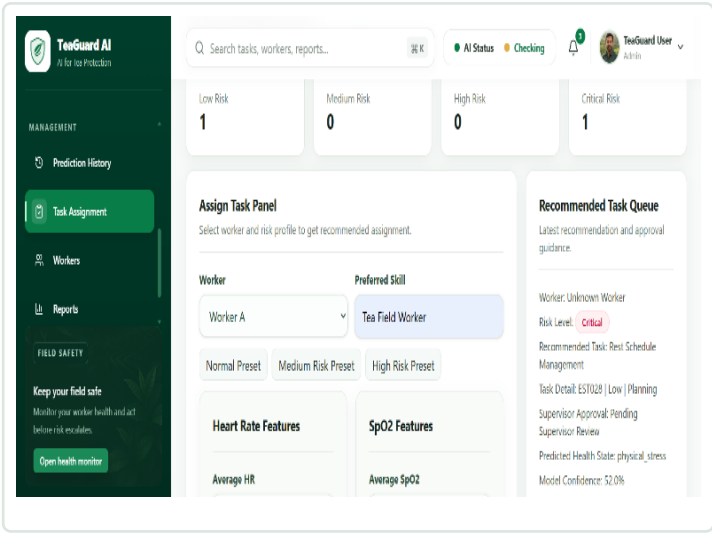


Results: 96.1% mAP@0.5, 95.2% precision, 93.8% recall, and 94.5% F1-score.
Limitation: Further validation is needed using new, unseen tea leaf images under real field conditions.

Labour Management & Welfare Optimization



- **Worker Monitoring:** Collects heart rate and SpO_2 data from wearables using MQTT.
- **Stress Classification:** Uses a LightGBM model to classify activity as Stress.
- **Task Recommendation:** Suggests tasks from a task catalogue for supervisor review.

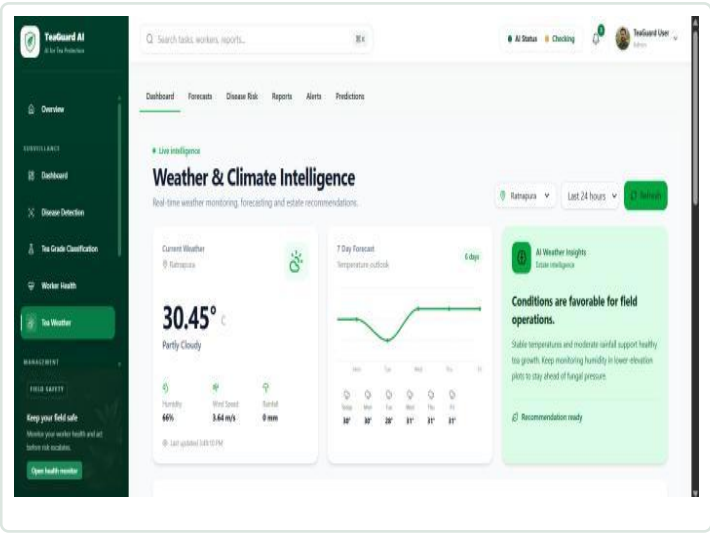


Limitation: Worker activity predictions depend on the accuracy and reliability of the collected sensor data.

Tea-Specific Weather & Climate Intelligence



- **Weather Data:** Collects live weather data every 5 minutes using OpenWeatherMap.
- **Prediction:** Uses Random Forest and Linear Regression models trained on 40177 historical records.
- **Dashboard:** Shows 8-hour forecasts, disease-risk levels, alerts, and recommended actions for Nuwara Eliya, Kandy, and Ratnapura.

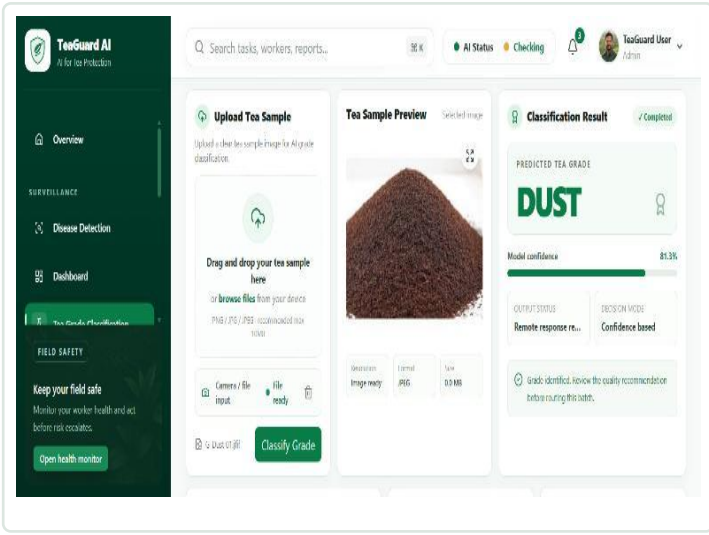


Limitation: The frontend connects directly to the weather service instead of using the API Gateway.

AI-Based Tea Sorting & Grading Automation



- **Image Classification:** Classifies processed-tea images using a deep learning model.
- **Five Grades:** BM, BOP, Broken Tea, Dust, Fanning 2, PF, and PW Dust.
- **Output:** Displays the predicted grade and confidence score.



Limitation: Software prototype only; no physical sorting conveyor or hardware is implemented.

User Feedback



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0770405358

Date - 10.09.2026.

To Whom It May Concern

Subject: Confirmation of Review of the Tea Guard AI Research System

This is to confirm that Wewewatte Estate Co Pvt Ltd, located at WewewatTE Estate Co (Pvt)Ltd, Rasagala Road, Balangoda was given an opportunity to review and evaluate the Tea Guard AI research system Developed by a student research team of the Sri Lanka Institute of Information Technology (SLIIT). The system is developed under the research project titled "Intelligent Decision Support System for Sri Lankan **Tea Estate Management"** (Project ID: R26-IT-118).

Based on the demonstration and review carried out by our representative on 10/09/2026, we have reviewed the system functions indicated below. The system aims to support tea estate and tea factory operations through digital decision-support capabilities. Our comments and confirmation are limited to the functions that were actually demonstrated or tested by our organization.

The research project includes the following components:

- **1. Tea Leaf Disease Detection and Pattern Analysis – (Miuranga W.A.R. - IT22209320)**
- **2. Labor Management and Welfare Optimization – (Gunasekara L.M.N.P. - IT22267986)**
- **3. Tea-Specific Weather and Climate Intelligence System – (Gunasekara G.N.D. IT22279620)**
- **4. AI-Based Tea Sorting and Grading Automation – (Pathirana I.M. - IT22282668)**

Following our review, we acknowledge that the system presents research based digital approaches that may be relevant to tea industry operations. The specific benefits, suitability, and any limitations identified during our reviews are stated below:

The system offers potential benefits in disease detection, labor management, weather monitoring, And Tea grading, supporting informed decision-making in the tea industry.

This letter is issued at the request of the research team as confirmation that our organization reviewed the System and provided feedback for academic research purposes. It does not constitute a guarantee of Commercial performance, formal certification, or confirmation of operational adoption unless separately Stated by our organization.

Yours faithfully,

Signature: _____

Name of authorized representative: H. M. D. S. Karunarathna

Designation: Manager

Wewawatte Estates Co. Ltd

Official stamp: _____

Manager

Limitations of the Overall System

- **Disease Detection:** 96.1% mAP@0.5, 95.2% precision, 93.8% recall, and 94.5% F1-score.
- **Labour Management:** 82% model accuracy.
- **Weather Intelligence:** 89% reported prediction accuracy.
- **Tea Grading:** 99.7% average precision, recall, and F1-score.
- **Future Improvements:** Expand datasets, improve model performance, integrate more IoT devices, and develop a mobile application